

# Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach



Seminar of the MathNet team (Inria Paris)

Louis HAUSEUX (Inria Sophia Antipolis, superv. by K. AVRACHENKOV & J. ZERUBIA)

Joint work with Nahuel SOPRANO-LOTO (Inria Paris)  
and Konstantin AVRACHENKOV (Inria Sophia Antipolis)

Inria Paris, Room Paul ERDÖS (A115) – 2 p.m. Monday 15 June 2026



# Outline

01

# Motivation & the Sankararaman–Baccelli Model



# Haplotype Assembly

## 1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

-----  
**Haplotype 1**  
(Truth  $\Sigma = +$ )

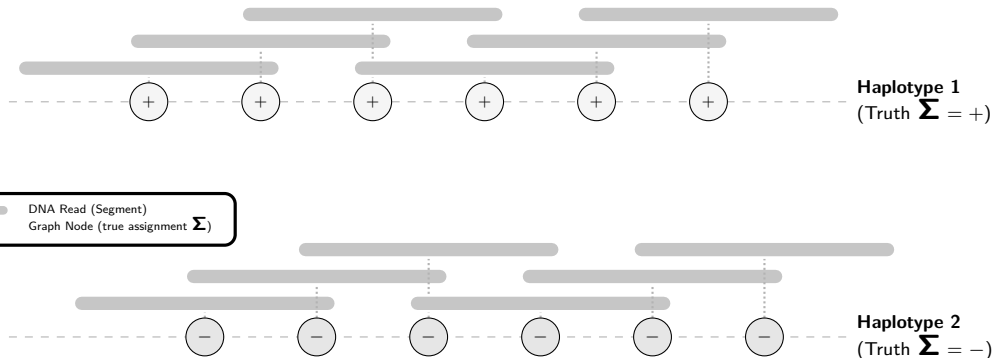
-----  
**Haplotype 2**  
(Truth  $\Sigma = -$ )



# Haplotype Assembly

## 2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.



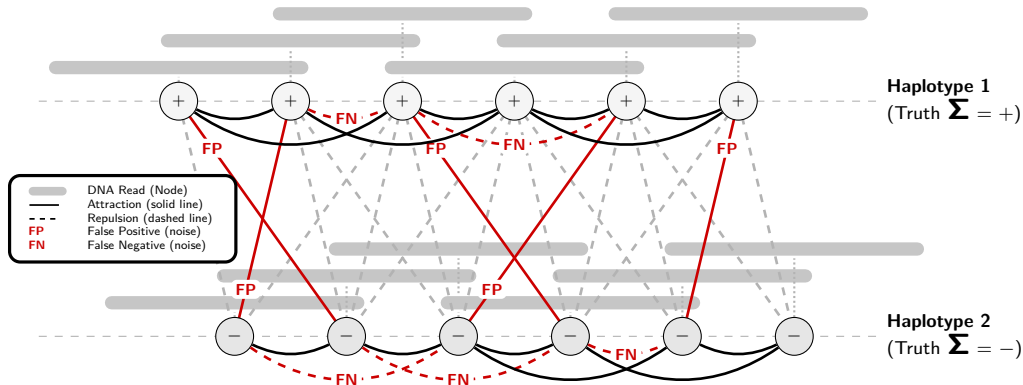


# Haplotype Assembly

## 3. Modeling as a Signed Graph

Nodes = reads      Edges = overlaps

**Goal:** Reconstruct the true haplotype assignments  $\Sigma$  from this observed signed graph.





## Motivation: Signed Community Detection

### Community Detection on Signed Graphs ( $K = 2$ )

- ▶ We aim to partition nodes into  $K = 2$  communities:  $\Sigma_i \in \{+1, -1\}$ .
- ▶ Observed links represent **attractions** (solid lines) and **repulsions** (dashed lines).
- ▶ Under the hood, this is a **Stochastic Block Model** (SBM) setup.

### The Geometric Challenge

- ▶ Nodes are embedded in space (e.g., genomic positions on  $\mathbb{Z}$  or  $\mathbb{R}^d$ ).
- ▶ Edge probabilities depend directly on the distance  $r = \|x_i - x_j\|$ :

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\text{in}}(r)$$

$$\mathbb{P}(\text{repulsive edge} \mid \text{different communities}) = f_{\text{out}}(r)$$

- ▶ **All the difficulty (and the interest!) comes from this geometric dependency.**



# The Sankararaman–Bacelli Model (GSBM)

## Model Definition [1]

- ▶ **Spatial Nodes**  $V_n$ : Hom. POISSON PP  $\mathcal{X}$  in  $\mathbb{R}^d$  [2] restrict. to a  $n$ -volume window.
- ▶ **Latent Communities**: Uniform assignments  $\Sigma_i \in \{+1, -1\}$  ( $K = 2$ ).
- ▶ **Geometric Overlaps**: Observed connection  $e = \{i, j\} \in H$  with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\text{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction),} \\ f_{\text{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion),} \end{cases}$$

with distance  $r_e = \|X_i - X_j\|$  and connecting functions  $f_{\text{in}}(r) \geq f_{\text{out}}(r)$ .

## Goal: Weak Recovery

- ▶ Find an algorithm  $\tau$  (independent of  $\Sigma$ ) reconstructing assignments:

$$\lim_{n \rightarrow \infty} \mathbb{E}[\text{ov}_n(\Sigma, \tau)] \geq \frac{1}{2} + \eta \quad \text{for some } \eta > 0,$$

using the **Community Overlap**:  $\text{ov}_n(\Sigma, \tau) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\tau_i = \pi(\Sigma_i)\}}.$





# Gibbs Model on Signed Graphs

## Attractive and Repulsive Edges

We partition the potential edges  $E = F \dot{\cup} \tilde{F}$ :

- ▶  $F$ : **Attractive** (ferromagnetic) edges, with weight  $W_e > 0$
- ▶  $\tilde{F}$ : **Repulsive** (antiferromagnetic) edges, with weight  $W_e < 0$

## Signed Gibbs Measure [3]

The probability of a label configuration  $\sigma \in \{+1, -1\}^{V_n}$  is  $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$ , where the energy  $U(\sigma)$  penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\} \in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\} \in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

The magnitudes  $|W_e|$  determine the strength of each constraint.



## From GSBM to the Gibbs Posterior

### Conditioning on the Point Cloud

- ▶ With fixed spatial positions  $X$ , all pairwise distances  $r_e = \|X_i - X_j\|$  are known.
- ▶ Under a uniform prior on  $\Sigma$ , the **posterior** distribution given the observed graph  $H$  is:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) \propto \mathbb{P}(H \mid X, \sigma)$$

### Log-Likelihood as Signed Gibbs Energy

- ▶ Choosing weights  $W_e(H)$  as below:

$$W_e(H) = \begin{cases} \log \left( \frac{f_{\text{in}}(r_e)}{f_{\text{out}}(r_e)} \right) > 0 & \text{if } e \in H \text{ (attractive } \in F), \\ -\log \left( \frac{1 - f_{\text{out}}(r_e)}{1 - f_{\text{in}}(r_e)} \right) < 0 & \text{if } e \notin H \text{ (repulsive } \in \tilde{F}). \end{cases}$$

- ▶ ... yields the exact correspondence:

$$\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$$



# The Percolation Obstruction

## Necessary Condition for Weak Recovery [1]

- ▶ Fragmented graphs (small, isolated components) prevent global reconstruction.
- ▶ Global information flow requires the graph to **percolate** [4] (possess a giant component).

## Theorem (Sankararaman & Baccelli [1])

*Independent bond percolation with edge probability*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

*yielding a giant component is a **necessary condition** for weak recovery.*

- ▶ **Our goal:** Retrieve and extend this result using unified **Bayesian MCMC coupling**.

02

# The Bayesian Framework & Coupling



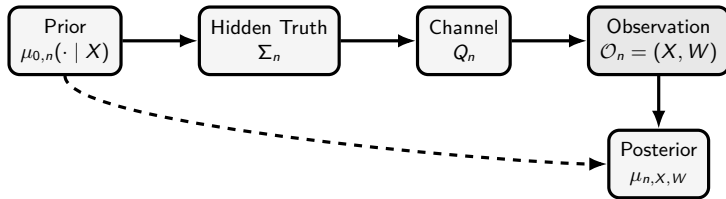


# The General Bayesian Model

## Definition

1. **Nodes, Edges, Communities:** Nodes  $V_n = \{1, \dots, n\}$ , edges  $E_n$ , communities  $\mathcal{L}_K = \{0, \dots, K-1\}$ , and set of configurations  $\sigma \in \mathcal{C}_n = (\mathcal{L}_K)^{V_n}$ .
2. **Features:**  $X_n \in \mathcal{X}_n$  according to distribution  $\nu_n$ .
3. **Prior on configurations:**  $\mu_{0,n}(\mathrm{d}\sigma \mid x)$  over  $\mathcal{C}_n$ .
4. **(Noisy) channel:**  $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$ .

The joint law is  $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x) \mu_{0,n}(\mathrm{d}\sigma \mid x) Q_n(\mathrm{d}w \mid x, \sigma)$ .





## Posterior Resampling (Nishimori Identity)

### Replacing Truth by a Replica

- ▶ Comparing the algorithm's output with the hidden truth  $\Sigma_n$  is equivalent to comparing it with a posterior replica  $\Sigma_n^{(1)}$ .

### Theorem (Posterior Resampling / Nishimori Identity [5])

Let  $\tau_n$  be any community detection algorithm. Let  $\Sigma_n^{(1)} \sim \mu_{n, X_n, W_n}$  be a replica drawn independently from the posterior. For any bounded test function  $\Psi_n$ :

$$\mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n, \tau_n)] = \mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n^{(1)}, \tau_n)] .$$

### Corollary (Overlap Identity)

For any algorithm  $\tau_n$  and threshold  $s \in [0, 1]$ :

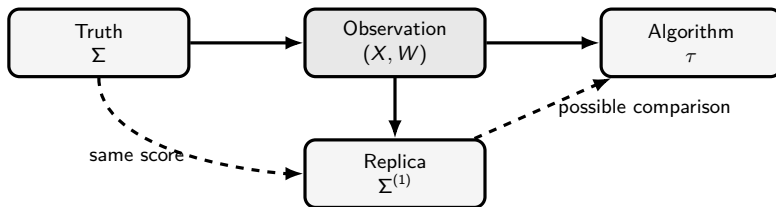
$$\mathbb{P} [\text{ov}_n(\Sigma_n, \tau_n) \geq s] = \mathbb{P} [\text{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq s] .$$



# Invariant Coupling

## Markov Chain Dynamics as a Source of Coupling

- ▶ We construct the posterior replica  $\Sigma^{(1)}$  starting from the hidden truth  $\Sigma$ .
- ▶ Running **exactly one step** of a posterior-invariant MARKOV kernel  $M$  initialized at  $\Sigma$  yields a valid replica  $\Sigma^{(1)} \sim M(\Sigma, \cdot)$ .



# 03

## MCMC Dynamics & Performance Bounds

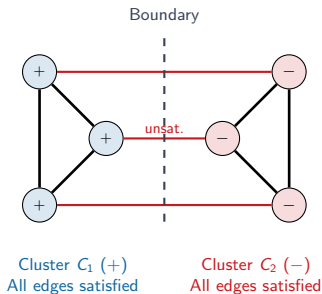




# Single-Spin Dynamics & Geometric Bottleneck

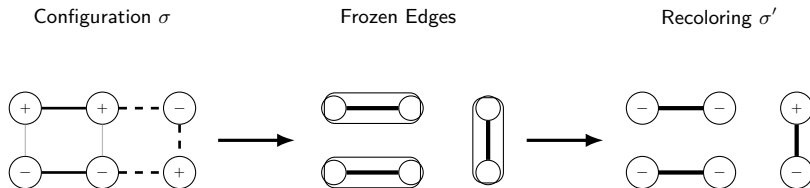
## Single-Spin Update Limitations

- ▶ **Geometry makes updates slow:** Single-spin dynamics (GLAUBER / METROPOLIS-HASTINGS [6]) update one node at a time.
- ▶ **Contradictory Clusters:** Two internally valid (highly satisfied) but mutually contradictory clusters are locked.
- ▶ **High Energy Barrier:** Flipping a single boundary node violates internal edges, creating a local energy barrier.
- ▶ **Conclusion:** Need for geometry  $\implies$  **Cluster Dynamics** (updating whole regions).





## One Step of Signed Swendsen–Wang [7, 8]



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.



# Swendsen–Wang Signed Cluster Dynamics

## Swendsen–Wang (SW) Signed Cluster Dynamics [7, 8]

- ▶ Update whole clusters of vertices simultaneously to speed up mixing.
- ▶ **Freezing Step:** For each edge  $e = \{i, j\}$ :
  - If  $e$  is satisfied ( $W_e > 0$  and  $\sigma_i = \sigma_j$ , or  $W_e < 0$  and  $\sigma_i \neq \sigma_j$ ), freeze it with probability:

$$p_e^{\text{frozen}} = 1 - e^{-|W_e|}.$$

- If  $e$  is unsatisfied, do not freeze it ( $p_e^{\text{frozen}} = 0$ ).
- ▶ **Recoloring Step:** Flip each frozen cluster independently with probability  $1/2$ .



# High-Order Triangular Dynamics

## The Frustration Obstruction

- ▶ **Frustration:**  $\nexists \sigma$  satisfying all edges (conflicting constraints).
- ▶ **Excessive Freezing:** In frustrated configurations, standard SW satisfies and freezes too many edges independently, blocking/freezing the MCMC dynamics.
- ▶ **Higher-Order Correlation:** Group interactions into **triangles** to correlate freezing and **freeze** as little as possible.

## Triangular Cluster Dynamics [9]

- ▶ **Energy Decomposition:** Distribute edge energy  $U$  over triangles:  $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$ .
- ▶ **Freezing Mechanism:** Apply a joint freezing mechanism on the triangles (rather than independent edges) to keep cluster sizes active.



## Triangular Dynamics Freezing Rules

| $\triangle$ <b>Attractive Triangle (Isotropic)</b> |             |          |          |          |                  |
|----------------------------------------------------|-------------|----------|----------|----------|------------------|
| Configuration                                      | $\emptyset$ | $\{ab\}$ | $\{ac\}$ | $\{bc\}$ | $\{ab, ac, bc\}$ |
| $(+, +, +)$                                        | $b_u$       | 0        | 0        | 0        | $a_u$            |
| Other                                              | 1           | 0        | 0        | 0        | 0                |

---

| $\triangleleft \triangle$ <b>Repulsive Triangle (Isotropic)</b> |             |          |          |          |                  |
|-----------------------------------------------------------------|-------------|----------|----------|----------|------------------|
| Configuration                                                   | $\emptyset$ | $\{ab\}$ | $\{ac\}$ | $\{bc\}$ | $\{ab, ac, bc\}$ |
| $(+, +, +)$                                                     | 1           | 0        | 0        | 0        | 0                |
| $(+, +, -)$                                                     | $b_u$       | 0        | $a_u/2$  | $a_u/2$  | 0                |
| $(+, -, +)$                                                     | $b_u$       | $a_u/2$  | 0        | $a_u/2$  | 0                |
| $(+, -, -)$                                                     | $b_u$       | $a_u/2$  | $a_u/2$  | 0        | 0                |

**Table:** Freezing rules for triangular dynamics ( $a_u = 1 - e^{-2u}$ ,  $b_u = e^{-2u}$ ), where  $u$  is the energy transferred by each edge.

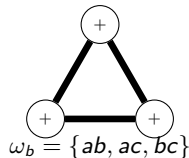
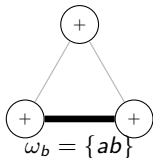
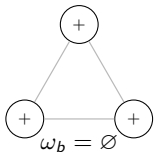


# General Swendsen–Wang Cluster Dynamics: Framework

## General Framework [10]

Suppose the Hamiltonian decomposes as:  $U(\sigma) = U_0(\sigma) + \sum_{b \in \mathcal{B}} U_b(\sigma)$ , where each  $b \in \mathcal{B}$  is a **bond** (e.g., edge, triangle, clique).

1. **Freezing Step:** For each bond  $b \in \mathcal{B}$ , freeze a sub-bond  $\omega_b \subseteq b$  with probability  $P_b^\sigma(\omega_b)$ .
2. **Recoloring Step:** Recolor each frozen cluster independently according to a distribution preserving the prior  $\mu_0(\sigma) \propto e^{-U_0(\sigma)}$ .





## General Swendsen–Wang Cluster Dynamics: Invariance

### Definition (Local Reversibility)

For two configurations  $\sigma, \sigma'$  and a bond  $b$ , we define the compatibility probability:

$$P_b^{\sigma \rightarrow \sigma'}(\omega_b) = \begin{cases} P_b^\sigma(\omega_b), & \text{if } P_b^{\sigma'}(\omega_b) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The freezing rule is **locally reversible** if for all  $b, \omega_b, \sigma, \sigma'$ :

$$P_b^{\sigma \rightarrow \sigma'}(\omega_b) = e^{-(U_b(\sigma') - U_b(\sigma))} P_b^{\sigma' \rightarrow \sigma}(\omega_b)$$

### Theorem (Gibbs Invariance [10])

If freezing choices are independent, locally reversible, and  $P_b^\sigma(\emptyset) > 0$  for all  $b, \sigma$ :

1. The global cluster dynamics is reversible with respect to  $\mu(\sigma) \propto e^{-U(\sigma)}$ .
2. The condition  $P_b^\sigma(\emptyset) > 0$  ensures the **irreducibility** of the MARKOV chain.

# Percolation as a Necessary Condition for Recovery

## Theorem (Percolation Bound on Overlap [10])

Under the uniform prior  $\mu_0$ , if frozen clusters are recolored independently and uniformly, then for the one-step posterior replica  $\Sigma_n^{(1)}$ , any algorithm  $\tau_n$ , and any  $\eta > 0$ :

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[ \text{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq \frac{1}{K} + \frac{K-1}{K}(\theta^{\max} + \eta) \right] = 0$$

where  $\theta^{\max}$  is the asymptotic fraction of nodes in giant frozen components.

## Corollary (Percolation is Necessary)

If there is no percolation ( $\theta^{\max} = 0$ ), the maximum overlap is bounded by  $1/K$  (random guess): weak recovery is impossible. Hence, **percolation is a necessary condition for weak recovery**.

## Key Intuition

Small clusters are recolored independently, washing out their contribution. Only giant (macroscopic) components can carry global community information.





## Coupling & the Sankararaman–Baccelli Bound

### One Step of Classical Edge-by-Edge Swendsen–Wang

Let's run one step of standard SWENDSEN–WANG (edges) initialized at the hidden truth  $\Sigma$ . For any edge  $e$ , the probability that it is frozen is:

► **Case 1** ( $\Sigma_i = \Sigma_j$ ):

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e) \left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e) \left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

► **Case 2** ( $\Sigma_i \neq \Sigma_j$ ):

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e)) \left(1 - e^{-|W_e|}\right) = (1 - f_{\text{out}}(r_e)) \left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

### Corollary (Retrieving the S&B Bound [1])

*Under the classical edge SW dynamics, the frozen graph is **exactly** an independent bond percolation with edge probability:*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$



## Strict Improvement of the S&B bound (on the triangular grid)

### Homogeneous Model ( $f_{\text{in}} = p$ , $f_{\text{out}} = 1 - p$ )

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- **Edge Dynamics (Standard SW):** Frozen graph is independent edge percolation with parameter  $p_{\text{edge}} = 2p - 1$ . Critical threshold:

$$p_c^{\text{edge}} = \frac{1}{2} + \sin\left(\frac{\pi}{18}\right) \approx 0.673\dots$$

- **Triangular Dynamics:** Frozen graph yields correlated triangle percolation. Critical threshold:

$$p_c^{\triangle} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192\dots$$

### Corollary (Strictly Better Impossibility Bound [10])

*Since  $p_c^{\text{edge}} < p_c^{\triangle}$ , for any  $p \in [0.673\dots, 0.7192\dots)$ , the standard S&B bound cannot conclude (edge dynamics percolate), whereas triangular dynamics do not, proving that weak recovery is impossible.*

# 04

## Open Questions & Conclusion





# Open Questions

## Limits of the One-Replica Coupling

- ▶ Our Bayesian framework combined with MCMC cluster dynamics allows us to construct **one** replica  $\Sigma_n^{(1)}$ .
- ▶ This only yields a bound (a **necessary condition** for weak recovery).

## Towards a Necessary and Sufficient Condition

- ▶ To achieve equality (a **necessary and sufficient condition**), a framework involving **two replicas** would be required.
- ▶ How can we define and analyze such a two-replica coupling under MCMC dynamics?

# Towards a Practical Community Detection Algorithm

## Limits of Direct Gibbs Sampling

- ▶ Empirically, recovering communities by sampling directly from the GIBBS distribution does not perform well.
- ▶ This is likely because the **mixing time** of the MARKOV chain is too large.

## Estimating the Correlation Matrix

- ▶ Constructing an estimator of the correlation matrix:

$$\mathbb{P}[\mathbb{I}_{\{\sigma_i = \sigma_j\}}]$$

gives significantly better results.

- ▶ **Future work:** It would be highly relevant to theoretically study why this approach is much more robust.



## Distance Adapted to Cluster Dynamics

### Hamming Similarity on the Quotient Space

- ▶ The standard overlap is essentially a **HAMMING** similarity on the configuration space quotiented by label permutations:  $\text{ov}_n(\sigma, \sigma') = \max_{\pi \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\sigma'_i = \pi(\sigma_i)\}}.$

### Markov Transition-Adapted Distance

- ▶ We should consider a distance that better reflects the transitions of the **MARKOV** cluster dynamics.
- ▶ For classical **SWENDSEN–WANG** (edges), this corresponds to the **symmetric difference** (the sum of edge weights satisfied by exactly one of the configurations):

$$d_{\text{sym}}(\sigma, \sigma') = \frac{1}{2} \sum_{e=\{i,j\}} |W_e| \left| \mathbb{I}_{\{\sigma \text{ satisfies } e\}} - \mathbb{I}_{\{\sigma' \text{ satisfies } e\}} \right|$$

- ▶ This would also establish a link to the **spectral analysis of the graph** (e.g., via Laplacian eigenvalues).

# Application: 3-SAT Solving via Signed Graphs

## The 3-SAT Problem

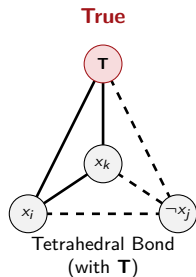
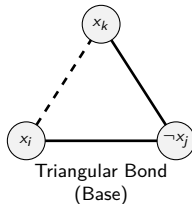
Find a truth assignment satisfying a set of clauses, e.g.,

$$\mathcal{C} = x_i \vee \neg x_j \vee x_k$$

## Clause Encoding

We introduce a reference node **T** (True):

- ▶ **Triangle** (literals  $x_i, \neg x_j, x_k$ ): edge signs opposite to the literal product.
- ▶ **Tetrahedron (with T)**: edge signs conforming to the literal signs.





## Bibliography I

- [1] Abishek Sankararaman and François Baccelli. “Community Detection on Euclidean Random Graphs”. In: *Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)* (2018), pp. 2181–2200. DOI: 10.1137/1.9781611975031.142.
- [2] François Baccelli and Bartłomiej Błaszczyszyn. *Stochastic Geometry and Wireless Networks, Volume I - Theory*. Vol. 1. Foundations and Trends in Networking. NoW Publishers, 2009, p. 150. DOI: 10.1561/13000000006.
- [3] Jörg Reichardt and Stefan Bornholdt. “Statistical mechanics of community detection”. In: *Phys. Rev. E* 74 (1 July 2006), p. 016110. DOI: 10.1103/PhysRevE.74.016110.
- [4] Mathew Penrose. *Random Geometric Graphs*. Vol. 5. Oxford Studies in Probability. Oxford University Press, 2003. DOI: 10.1093/acprof:oso/9780198506263.001.0001.
- [5] Hidetoshi Nishimori. “Exact results and critical properties of the Ising model with competing interactions”. In: *Journal of Physics C: Solid State Physics* 13.21 (1980), pp. 4071–4076. DOI: 10.1088/0022-3719/13/21/012.
- [6] Persi Diaconis. “The Markov chain Monte Carlo revolution”. In: *Bulletin of the American Mathematical Society* 46.2 (2009), pp. 179–205. DOI: 10.1090/S0273-0979-08-01238-X.



## Bibliography II

- [7] Robert H. Swendsen and Jian-Sheng Wang. “Nonuniversal critical dynamics in Monte Carlo simulations”. In: *Phys. Rev. Lett.* 58 (2 Jan. 1987), pp. 86–88. DOI: [10.1103/PhysRevLett.58.86](https://doi.org/10.1103/PhysRevLett.58.86).
- [8] Robert G. Edwards and Alan D. Sokal. “Generalization of the Fortuin-Kasteleyn-Swendsen-Wang representation and Monte Carlo algorithm”. In: *Phys. Rev. D* 38 (6 Sept. 1988), pp. 2009–2012. DOI: [10.1103/PhysRevD.38.2009](https://doi.org/10.1103/PhysRevD.38.2009).
- [9] Louis Hauseux, Nahuel Soprano-Loto, and Konstantin Avrachenkov. “Higher-order Monte Carlo cluster dynamics for community detection in Euclidean graphs”. In: *Asilomar Conference on Signals, Systems, and Computers 2025*. Pacific Grove (CA), United States, Oct. 2025. URL: <https://inria.hal.science/hal-05267074>.
- [10] Louis Hauseux, Nahuel Soprano-Loto, and Konstantin Avrachenkov. *A Bayesian Framework for Community Detection on Signed Graphs. Tighter Bounds with Higher-Order Interaction Markov Chain Monte Carlo Cluster Dynamics*. Article en cours de rédaction. 2026+.

*Merci.*

